Coffee reduces risk for hepatocellular carcinoma: an updated meta-analysis.

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BACKGROUND & AIMS: Coffee consumption has been suggested to reduce the risk for hepatocellular carcinoma (HCC). We performed a meta-analysis of epidemiological studies to provide updated information on how coffee drinking affects HCC risk. METHODS: We performed a PubMed/MEDLINE search of the original articles published in English from 1966 through September 2012, on case-control or cohort studies that associated coffee consumption with liver cancer or HCC. We calculated the summary relative risk (RR) for any, low, and high consumption of coffee vs no consumption. The cut-off point for low vs high consumption was set to 3 cups per day in 9 studies and 1 cup per day in 5 studies. RESULTS: The summary RR for any coffee consumption vs no consumption was 0.60 from 16 studies, comprising a total of 3153 HCC cases (95% confidence interval [CI], 0.50-0.71); the RRs were 0.56 from 8 case-control studies (95% CI, 0.42-0.75) and 0.64 from 8 cohort studies (95% CI, 0.52-0.78). Compared with no coffee consumption, the summary RR was 0.72 (95% CI, 0.61-0.84) for low consumption and 0.44 (95% CI, 0.39-0.50) for high consumption. The summary RR was 0.80 (95% CI, 0.77-0.84) for an increment of 1 cup of coffee per day. The inverse relationship between coffee and HCC risk was consistent regardless of the subjects' sex, alcohol drinking, or history of hepatitis or liver disease. CONCLUSIONS: From this meta-analysis, the risk of HCC is reduced by 40% for any coffee consumption vs no consumption. The inverse association might partly or largely exist because patients with liver and digestive diseases reduce their coffee intake. However, coffee has been shown to affect liver enzymes and development of cirrhosis, and therefore could protect against liver carcinogenesis.